TEAM## · HELIN SAYGILI · NISANUR KARA · CEVAHIR ATICI · EMRE KILIÇ

# Smart Campus Announcement Assistant

**RESEARCH QUESTION**

**Does fine-tuning BERTurk classify campus announcements more accurately (Macro-F1) than a TF-IDF + logistic-regression baseline, on announcements written outside our generation templates?**

**WHAT A "NO" LOOKS LIKE**

BERTurk gains **< 0.03 Macro-F1**, or the paired-bootstrap 95% CI of the gain includes 0 → a linear model is enough for stage 01, and the project's effort should go to extraction and QA.

**CHANGED SINCE THE PROPOSAL — AND WHY**

**Old:** "How can NLP techniques turn scattered university announcements into structured, searchable information and answer students' questions?"
**Why:** an open "how" question has no "no" result, and on our template-based test set TF-IDF already scores **1.000**. We kept the system and narrowed the question to the one comparison that is still open: out-of-template text.

DATA AS LOADED

# 420 labelled announcements + 24 hand-written, test set frozen

| Set | Source | N | TR / EN |
|---|---|---|---|
| Train | Synthetic “campus world” scripts/generate_dataset.py, seed 42 | **252** | 198 / 54 |
| Dev | | **84** | 66 / 18 |
| **Test A (frozen)** | | **84** | **71 / 13** |
| **Test B (frozen)** | **Hand-written, real-style, never trained on** | **24** | **19 / 5** |
| GTÜ raw (unlabelled) | gtu.edu.tr, 24 departments, collected Nov 11 scripts/collect_gtu.py | **818** | ≈ all TR |

6 classes: exam 92 · event 76 · registration 71 · cancellation 63 · room change 59 · scholarship 59. Split: stratified 60/20/20, seed 2026, ids in `data/splits/midproject_split.json` (sha256 `2cafd21d7400`). Preprocessing: 420 → clean_text 420 → dedup 420 (0 removed). 1,292 entity spans (DATE/TIME/LOCATION/COURSE).

**Licence:** all text is team-written (generator + hand-written set) → released CC BY 4.0; GTÜ announcements are public pages © GTÜ: we release only URLs + our labels, not the text.

### ONE REAL EXAMPLE (ANN-0001, TRAIN)

**Title:** KIM107 dersi iptal
**Text:** “1 Kasım 2026 tarihinde yapılması planlanan KIM107 Genel Kimya dersi, öğretim üyesinin rahatsızlığı nedeniyle yapılmayacaktır.”

**category** = cancellation
**entities** = DATE “1 Kasım 2026” → 2026-11-01 · COURSE “KIM107” → KIM 107
**summary** = “KIM107 dersi 1 Kasım tarihinde yapılmayacak.”

### HONEST CAVEAT

Train/dev/test A are synthetic, so they share vocabulary with our rules and templates. That is why Test B exists and why a real test set (Test C) is the biggest item on the plan.

BASELINE RESULT

# Test A is saturated; out-of-template text is where the gap is

| System (trained on train, 252) | Dev Macro-F1 | Test A Macro-F1 [95% CI] | Test A Acc | Test B Macro-F1 [95% CI] | Test B Acc |
|---|---|---|---|---|---|
| Majority class | 0.059 | 0.059 [0.04, 0.08] | 0.214 | 0.048 [0.01, 0.09] | 0.167 |
| Keyword rules | 0.851 | 0.840 [0.76, 0.91] | 0.857 | 0.801 [0.60, 0.94] | 0.792 |
| TF-IDF + NB | 1.000 | 1.000 [1.00, 1.00] | 1.000 | 0.789 [0.56, 0.94] | 0.792 |
| TF-IDF + linear SVM | 1.000 | 1.000 [1.00, 1.00] | 1.000 | 0.874 [0.68, 1.00] | 0.875 |
| **TF-IDF + LogReg (main baseline)** | **1.000** | **1.000** [1.00, 1.00] | **1.000** | **0.874** [0.68, 1.00] | **0.875** |

**0.874**

TF-IDF + LogReg on hand-written Test B (n = 24)

**0.126**

drop from Test A to Test B

Spread: all baselines are deterministic, so the interval is a **95% bootstrap CI** (1,000 resamples of the test set, seed 0). Test B's CI is ~0.3 wide: with 24 items we cannot resolve a 0.03 difference → we need a larger real test set.
Reproduce: `python evaluation/midproject.py`

ERROR ANALYSIS

# 65 baseline errors in 7 categories

| # | Category | N | Example (gold → predicted) |
|---|---|---|---|
| **E1** | Topic beats action<br>exam room change → exam | **21** | *"27 Kasım 2026 tarihinde yapılacak CENG 213 mazeret sınavı Kütüphane Seminer Salonu yerine Kültür Merkezi salonunda ya…"* **room_change → exam** |
| **E4** | No keyword fires<br>falls back to a default class | **16** | *"The double major applications period starts on November 18, 2026 and ends on November 22, 2026. All steps are complet…"* **registration → exam** |
| **E2** | Shared cue "başvuru"<br>registration → scholarship | **13** | *"Akademik takvim gereği çift anadal başvuruları 18.11.2026 - 22.11.2026 tarihleri arasında yapılacaktır. Süresi içinde…"* **registration → scholarship** |
| **E3** | Registration about an exam<br>"exam applications" → exam | **7** | *"The English proficiency exam applications period starts on November 1, 2026 and ends on 5 November 2026. All steps ar…"* **registration → exam** |
| **E5** | Event mentions exams<br>"sınav haftası öncesi …" → exam | **3** | *"Sınav haftası öncesi moral ve motivasyon için Doğa Yürüyüşü düzenliyoruz. Tarih: 17/11/2026, Saat: 15:00, Yer: Amfi 1…"* **event → exam** |
| **E6** | Room name = event cue<br>"Seminer Salonu" → event | **2** | *"Kapasite nedeniyle CENG 301 dersi 17.11.2026 itibarıyla Kütüphane Seminer Salonu dersliğinden D Blok 105 dersliğine t…"* **room_change → event** |
| **E7** | TF-IDF: unseen wording<br>hand-written set only | **3** | *"Join our guest lecture on large language models for Turkish on Dec 3 at 4 pm in Conference Hall. Everyone is welcome."* **event → cancellation** |

Keyword rules: 45 train + 12 dev + 5 Test B errors (rules are not trained, so train errors are fair); TF-IDF: 3 Test B errors. Test A errors were not inspected (frozen). File: `reports/midproject_errors.csv`

ERROR ANALYSIS

# What the errors tell us

### 1 · THE LABEL IS THE ACTION, NOT THE TOPIC (E1, E3, E5 = 31)

“MAT 102 sınavı Konferans Salonu yerine E-205'te” is a room change, but every cue word says exam. Bag-of-words counts cannot weigh “X yerine Y” against “sınav”. A contextual encoder can — this is the strongest argument for BERTurk.

### 2 · VOCABULARY GAPS (E4, E7 = 19)

“Don't miss…”, “no lecture”, “ekle-sil”, “guest lecture”, “tuition waiver”, “ilişik kesme”: the hand-written set uses words that never appear in training. TF-IDF fails here; pretrained subword knowledge may not.

### 3 · AMBIGUOUS SHARED CUES (E2, E6 = 15)

“başvuru” is both scholarship and registration; “Seminer Salonu” is a room, not a seminar. TF-IDF learns these from data (0 errors on Test A) — so these are a rules problem, not a learning problem.

### 4 · EXTRACTION (STAGE 02) ON TEST B: 7 ERRORS

Date ranges without a year (“15-17 Mayıs” → wrong year, 4), free-form places (“Fen Fakültesi amfileri”, 2), dotted time “17.30'da” (1). Micro-F1 is still 0.947.

### SO

Expect BERTurk to help on categories 1 and 2 only, which exist mainly outside the templates → measure on Test B and a real Test C, not on Test A.

METHOD STATUS

# BERTurk fine-tuning implemented and run on CPU

### THE ONE CONTROLLED CHANGE

Replace **TF-IDF features + LogReg** with a **fine-tuned BERTurk encoder** ( `dbmdz/bert-base-turkish-cased` ). Same split, same input (title + text), same labels, same metric.

### AS IMPLEMENTED

4 epochs · lr 3e-5 · batch 16 · max length 128 · weight decay 0.01 · seeds 13, 42, 2026 · `python evaluation/midproject.py --with-berturk`
Compute: 5.8 min per seed on a laptop CPU; final runs (BERTurk + XLM-R, 3 seeds, larger train) ≈ 2 GPU-hours on Colab T4.

| Dev set (n = 84) | Macro-F1 | Accuracy |
|---|---|---|
| Keyword rules | 0.851 | 0.857 |
| TF-IDF + LogReg | 1.000 | 1.000 |
| **BERTurk (mean ± std, 3 seeds)** | **0.807 ± 0.134** | **0.841 ± 0.100** |

**First finding:** BERTurk is **below** TF-IDF on dev and unstable across seeds (per seed: 0.918, 0.618, 0.885). Training loss is still 1.1–1.6 after 64 steps → under-trained, not a ceiling effect. Next: 10–20 epochs, lr 5e-5, epochs chosen on dev, on GPU. Test sets untouched until Dec 9.

### REST OF THE PIPELINE (ALREADY WORKING)

Extraction rules: micro-F1 0.947 on Test B · Summaries: structured template ROUGE-L 0.953 vs lead-1 0.432 · QA (50 questions, end-to-end with predicted labels): 50/50 answers correct · Flask web app + CLI.

PLAN TO THE FINAL

# Remaining experiments, owners, dates

| Step | Owner | Date |
|---|---|---|
| ~~Collect ≥ 300 real announcements~~ **done: 818 from 24 GTÜ departments** | Helin S. | Nov 11 ✓ |
| Double-annotate 150 GTÜ items (+ "other" class) → **Test C**, κ on 50, freeze | Helin S. + Nisanur K. | Nov 27 |
| BERTurk, 3 seeds on GPU, choose epochs on dev | Cevahir A. | Nov 24 |
| Ablation: XLM-R vs BERTurk; title removed | Cevahir A. | Dec 4 |
| Fix extraction errors X1–X3; hybrid NER on real data | Nisanur K. | Dec 4 |
| QA on 50 real student questions | Emre K. | Dec 7 |
| **One** test run of all systems on A/B/C + paired bootstrap | Cevahir A. | Dec 9 |
| Error analysis of the method | Nisanur K. | Dec 11 |
| Final report draft (paper format) + demo | Emre K. (all) | Dec 18 |

### PROTOCOL (FIXED TODAY, BEFORE METHOD RESULTS)

Macro-F1 (primary), accuracy · frozen tests A, B (+C on Nov 27) · 3 seeds for neural models · 95% bootstrap CI · **a gain is real only if ΔMacro-F1 ≥ 0.03 and the paired-bootstrap CI excludes 0**.

### BIGGEST RISK → FALLBACK

Real GTÜ text is far from our labels: median classifier confidence 0.89 on Test A vs **0.37 on the 818 GTÜ items**, and many are none of our 6 classes. Labelling may be slow / κ low.
**Fallback:** label 150 instead of 300, add an "other" class, report the minimum detectable difference.

### PROPOSAL FEEDBACK

RQ narrowed to a testable yes/no; baselines + error analysis added before any neural model.

APPENDIX A · PROJECT CARD 1/4

# Fields 1–3

## 1 · TITLE AND ABSTRACT

**Working title:** Do We Need BERTurk to Sort Campus Announcements? Linear Baselines vs. Fine-tuned Encoders on Out-of-Template Turkish Notices

**Draft abstract (116 words):** University announcements arrive through websites, e-mail and club channels, and students miss exam, room and deadline changes. We build a four-stage assistant (classification, information extraction, summarisation, cited question answering) for Turkish and English announcements and ask whether a fine-tuned Turkish encoder is needed for the first stage. On 420 labelled synthetic announcements a TF-IDF + logistic-regression baseline reaches 1.000 Macro-F1, but only 0.874 on 24 hand-written announcements; most baseline errors confuse the topic of an announcement with the action it asks for. Fine-tuned BERTurk reaches **[GAP: Test B / Test C Macro-F1 ± CI]** , which **[GAP: does / does not]** exceed our 0.03 threshold. The extraction rules reach 0.947 micro-F1 and end-to-end QA answers 50/50 questions.

## 2 · RESEARCH QUESTION

**RQ:** Does fine-tuning BERTurk classify campus announcements more accurately (Macro-F1) than TF-IDF + logistic regression on announcements written outside our generation templates?

**"No" result:** ΔMacro-F1 < 0.03 or the paired-bootstrap 95% CI includes 0 → the linear model suffices.

**Changed:** proposal asked "How can NLP techniques turn scattered announcements into structured, searchable information and answer questions?" — no falsifiable outcome, and the in-template test is saturated (TF-IDF 1.000). The system is unchanged; the question is narrowed to stage 01 on out-of-template text.

## 3 · CONTRIBUTIONS (TESTABLE CLAIMS)

- **C1.** On out-of-template announcements (Test B/C), fine-tuned BERTurk beats TF-IDF + LogReg by ≥ 0.03 Macro-F1 (paired bootstrap). Refuted if the CI includes 0.
- **C2.** The gain comes from "topic-vs-action" errors (E1/E3/E5): BERTurk removes ≥ 50% of them while E2/E6 errors are already 0 for TF-IDF.
- **C3.** A Turkish-only encoder (BERTurk) is at least as good as multilingual XLM-R on the 20% English announcements (ablation). Refuted if XLM-R is ≥ 0.03 better on English items.

# Fields 4–6

## 4 · RELATED WORK


- Devlin, J., Chang, M.-W., Lee, K., Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL-HLT*. — We fine-tune a BERT-style encoder on a small 6-class domain task, not on general benchmarks.
- Schweter, S. (2020). BERTurk – BERT models for Turkish. Zenodo, doi:10.5281/zenodo.3770924. — Our method model; we test whether its gains hold on short, mixed TR/EN campus notices.
- Conneau, A., Khandelwal, K., Goyal, N., et al. (2020). Unsupervised Cross-lingual Representation Learning at Scale. *ACL*. — XLM-R is our ablation for the English share of announcements.
- Wang, S., Manning, C. D. (2012). Baselines and Bigrams: Simple, Good Sentiment and Topic Classification. *ACL*. — Motivates a strong linear baseline; we add character n-grams for Turkish suffixes.
- Joulin, A., Grave, E., Bojanowski, P., Mikolov, T. (2017). Bag of Tricks for Efficient Text Classification. *EACL*. — Same claim that linear models are competitive; we test it on low-resource Turkish text.
- Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*. — Our QA retrieves and filters structured fields and cites sources instead of training a generator.
- Dror, R., Baumer, G., Shlomov, S., Reichart, R. (2018). The Hitchhiker's Guide to Testing Statistical Significance in NLP. *ACL*. — Basis for our paired-bootstrap decision rule.
- Dodge, J., Ilharco, G., Schwartz, R., et al. (2020). Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping. arXiv:2002.06305. — Why we report 3 seeds for every neural model.


## 5 · DATA

**Source:** synthetic "campus world" generator ( `scripts/generate_dataset.py` , seed 42, 4 source types) + 24 hand-written real-style announcements (team-authored, CC BY 4.0) + **818 raw GTÜ announcements** from 24 departments ( `scripts/collect_gtu.py` , public pages © GTÜ → we share URLs + labels only).

**Counts:** generated 420 → clean_text 420 → dedup 420 → split train 252 / dev 84 / test A 84; Test B 24. 335 TR / 85 EN; 1,292 entity spans. GTÜ: 818 collected (517 pages, 259 PDF titles, 42 links; 0 duplicates, 3 timeouts) → 150 to be labelled as Test C.

**Example:** "1 Kasım 2026 tarihinde yapılması planlanan KIM107 Genel Kimya dersi … yapılmayacaktır." → cancellation; DATE 2026-11-01, COURSE KIM 107.

**Guideline:** `docs/annotation_guidelines.md` (label = the student's required action). **Agreement:** not measured yet — κ on a 50-item double-annotated sample of the real set, due Nov 27.

## 6 · METHOD AND BASELINE

**Baseline as run:** TF-IDF (word 1–2-grams + char 3–5-grams, folded Turkish) + logistic regression (C = 5); also keyword rules, NB, SVM, majority.

**Method as implemented:** BERTurk ( `dbmdz/bert-base-turkish-cased` ) fine-tuned, 4 epochs, lr 3e-5, batch 16, max length 128; dev Macro-F1 0.807 ± 0.134 (3 seeds) vs 1.000 for TF-IDF — under-trained after 64 steps, so epochs will be tuned on dev.

**The one change:** the text representation + classifier (sparse TF-IDF + LogReg → fine-tuned encoder); data, input fields, split and metric are identical.

**Compute:** baselines < 10 s CPU; BERTurk 5.8 min/seed on CPU. Needed: ≈ 2 GPU-hours (Colab T4) for BERTurk + XLM-R × 3 seeds on the larger train set.

# Fields 7–9

## 7 · EVALUATION PROTOCOL (FIXED NOV 11, BEFORE METHOD RESULTS)

**Metric:** Macro-F1 over 6 classes (primary); accuracy (secondary). Extraction: micro P/R/F1 on (label, normalised value).

**Frozen tests:** A = 84 synthetic (ids + sha256 2cafd21d7400 in `data/splits/`); B = 24 hand-written; C = 150 real, frozen on Nov 27 before any model sees it. Each system is run on the tests once (Dec 9).

**Spread:** 3 seeds (13, 42, 2026) for neural models, mean ± std; 95% bootstrap CI (1,000 resamples, seed 0) for all.

**Smallest real difference:** ΔMacro-F1 ≥ 0.03 *and* paired-bootstrap 95% CI of Δ excludes 0.

## 8 · RESULTS SO FAR (FINAL RESULTS TABLE)

| System | Test A Macro-F1 [CI] | Test B Macro-F1 [CI] | Test C |
|---|---|---|---|
| Majority class | 0.059 [0.04, 0.08] | 0.048 [0.01, 0.09] | *Dec 9* |
| Keyword rules | 0.840 [0.76, 0.91] | 0.801 [0.60, 0.94] | *Dec 9* |
| TF-IDF + NB | 1.000 [1.00, 1.00] | 0.789 [0.56, 0.94] | *Dec 9* |
| TF-IDF + linear SVM | 1.000 [1.00, 1.00] | 0.874 [0.68, 1.00] | *Dec 9* |
| **TF-IDF + LogReg (main)** | **1.000 [1.00, 1.00]** | **0.874 [0.68, 1.00]** | *Dec 9* |
| BERTurk (3 seeds) | *Dec 9* | *Dec 9* | *Dec 9* |
| XLM-R base (ablation) | *Dec 9* | *Dec 9* | *Dec 9* |
| BERTurk, no title (ablation) | *Dec 9* | *Dec 9* | *Dec 9* |

Baselines are deterministic, so the CI comes from bootstrap; seed std applies to neural rows.

## 9 · ERROR ANALYSIS — 65 BASELINE CLASSIFICATION ERRORS (+ 7 EXTRACTION ERRORS)

| # | Category | N | Example (gold → predicted) |
|---|---|---|---|
| E1 | Topic beats action | **21** | *"27 Kasım 2026 tarihinde yapılacak CENG 213 mazeret sınavı Kütüphane Seminer Salonu yerine Kü…"* **room_change → exam** |
| E4 | No keyword fires | **16** | *"The double major applications period starts on November 18, 2026 and ends on November 22, 20…"* **registration → exam** |
| E2 | Shared cue "başvuru" | **13** | *"Akademik takvim gereği çift anadal başvuruları 18.11.2026 - 22.11.2026 tarihleri arasında ya…"* **registration → scholarship** |
| E3 | Registration about an exam | **7** | *"The English proficiency exam applications period starts on November 1, 2026 and ends on 5 No…"* **registration → exam** |
| E5 | Event mentions exams | **3** | *"Sınav haftası öncesi moral ve motivasyon için Doğa Yürüyüşü düzenliyoruz. Tarih: 17/11/2026,…"* **event → exam** |
| E6 | Room name = event cue | **2** | *"Kapasite nedeniyle CENG 301 dersi 17.11.2026 itibarıyla Kütüphane Seminer Salonu dersliğinde…"* **room_change → event** |
| E7 | TF-IDF: unseen wording | **3** | *"Join our guest lecture on large language models for Turkish on Dec 3 at 4 pm in Conference H…"* **event → cancellation** |
| X1–3 | Extraction: year-less date range / free-form place / "17.30'da" | **4/2/1** | *"konserler 15-17 Mayıs tarihlerinde Kampüs Açık Hava Tiyatrosu'nda"* → 2026 instead of 2027, place missed |

APPENDIX D · PROJECT CARD 4/4

# Fields 10–12

## 10 · RESEARCH LEAD SECTION — NISANUR KARA (RESEARCH LEAD)

**Uludoğan, G., Yirmibeşoğlu Balal, Z., Akkurt, F., Türker, M., Güngör, O., Üsküdarlı, S. (2024). TURNA: A Turkish Encoder-Decoder Language Model for Enhanced Understanding and Generation.** ***Findings of ACL 2024*****.** — Evaluates Turkish models on general benchmarks and finds encoder models such as BERTurk strong on classification; we test whether that holds for a tiny, noisy, mixed-language domain where a linear model is already near its ceiling.

**Edwards, A., Camacho-Collados, J. (2024). Language Models for Text Classification: Is In-Context Learning Enough?** ***LREC-COLING 2024*****.** — Shows fine-tuned smaller models often beat prompted LLMs on classification (English); we ask the prior question, whether fine-tuning beats a linear model at all, in Turkish, and can add a prompted Claude row as an extra comparison.

**Ablation we will run:** (a) XLM-R base vs BERTurk with the identical recipe (published multilingual baseline); (b) BERTurk without the title field; (c) learning curve with 25 / 50 / 100% of train. All on frozen tests A/B/C, 3 seeds.

## 11 · PLAN AND RISK

Real data: done Nov 11 (818 GTÜ items) · Test C double annotation + κ, freeze — Helin S. + Nisanur K. Nov 27 · BERTurk GPU runs — Cevahir A. Nov 24 · XLM-R / no-title ablation — Cevahir A. Dec 4 · extraction fixes + hybrid NER — Nisanur K. Dec 4 · QA on 50 real questions — Emre K. Dec 7 · single test run + paired bootstrap — Cevahir A. Dec 9 · method error analysis — Nisanur K. Dec 11 · report draft — Emre K. (all) Dec 18.

**Biggest risk:** real GTÜ text is far from our label set (median confidence 0.37 vs 0.89 on Test A; many items fit none of the 6 classes), so labelling is slow or κ is low. **Fallback:** 150 labelled items, an added "other" class, and the minimum detectable difference reported as a limitation.

**Proposal feedback:** RQ made falsifiable with an explicit "no"; baselines and error analysis added first.

## 12 · REPRODUCIBILITY

**Repository:** https://github.com/<team>/smart-campus-assistant

**One command (field 8):** `python evaluation/midproject.py` (add `--with-berturk` for the method dev row; `python evaluation/error_analysis.py` for field 9).

**Seeds:** data 42 · split 2026 · bootstrap 0 · neural 13 / 42 / 2026. **Versions:** Python 3.12.6, scikit-learn 1.9.1, numpy 2.5.4, torch 2.15.0 (CPU), transformers 5.22.2.

**Who did what:** Helin S. — data collection, guideline; Nisanur K. — research lead (related work, ablation design), extraction, error analysis; Cevahir A. — classifiers, evaluation protocol, repository; Emre K. — summarisation, QA, web demo.

**AI tools:** Claude (Anthropic) via Claude Code helped write evaluation scripts and draft slide text; every number was produced by running the scripts above and checked by the team.